# A Thermodynamic Description of Closure Tendencies and Points of Intervention

**Child Paper 4 — English Full Text v1.0 (Complete Academic Version)**

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## Abstract

This paper proposes a descriptive framework, rather than a strict application of physical law, for understanding why difference-preserving arrangements tend to be displaced by difference-reducing ones under specific institutional cost structures. Using thermodynamic analogy as descriptive scaffolding, it reorganizes the variables developed in this series — difference-retention (D), evaluative diversity (Var(φ)), and effective recomputation energy (E_r_eff) — into a map of closure tendencies and points of intervention across individual, institutional, and civilizational scales.

A key distinction is drawn between *closure* and *homogenization*. Closure refers to the deterioration of input conditions, that is, the decreasing availability of difference inputs capable of disrupting existing frames. Homogenization is its downstream consequence: the contraction of evaluative diversity. The thermodynamic analogy is applied specifically to this upstream condition — not to social disorder in general, but to the structural tendency for difference-preserving arrangements to carry higher costs than difference-eliminating ones under prevailing institutional designs.

The paper maps points of intervention at three scales. At the individual scale, closure corresponds to the depletion of E_r_eff described in Child Paper 2. At the institutional scale, it corresponds to the evaluative simplification described in Child Paper 5. At the civilizational scale, the paper introduces *civilizational Region B*: a condition in which aggregate output and functional stability are maintained while the capacity for frame-level renewal is collectively diminished. This is offered as an analytical construct, not an ontological claim about civilization as a unified subject.

The paper concludes that intervention must target not only the supply of difference, but also the institutional cost structures that make closure the default trajectory.

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## A note on the theoretical frameworks

OMD (Organizational Metastability Dynamics), RD (Recomputation Dynamics), and RLAF (Reinforcement Learning from AI Feedback) are theoretical frameworks developed within the present research program and are introduced here as analytical constructs rather than references to previously published models under those names.

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## 1. Introduction

### 1.1 The Problem

This paper addresses two questions.

The first: **why do institutions and civilizations tend, in the absence of intervention, to move in a direction that reduces difference?**

The second: **given that tendency, where can intervention preserve D (difference-retention) and recomputability?**

What is important here is that this paper does not claim that cessation is natural or inevitable. What it seeks to describe are the conditions under which, given a specific cost structure, a system tends toward closure. The subject of this paper is therefore not necessity but **closure tendency and the possibility of intervention**.

Child Paper 2 formalized the minimal causal chain from the decline of D to the cessation of M (recomputation capacity), T_eval (evaluative autonomy), and T_val (value autonomy). Child Paper 5 described the cycle in which that cessation is reproduced by institutional design. The role of this paper is not to reduce those two accounts to physical law, but to provide **a schematic map of why closure is readily selected as the lower-cost option, and where the points of intervention for design can be located**.

The purpose of this paper is therefore not to introduce new causal laws through thermodynamic vocabulary, but to redescribe the already formalized variable system of subject-injury as a schematic map of closure tendency and the possibility of intervention.

The “possibility of intervention” in this paper refers not to the negation of logical necessity, but to a conditional possibility: that if the historical and institutional conditions change, the strength of the closure tendency can also change. This paper adopts this sense of possibility as descriptive scaffolding — positioned between “it is not fate” and “anything can be changed.”

### 1.2 The Position of This Paper

This paper does not claim that thermodynamic laws can be strictly applied to society. What this paper claims is **structural analogy** and **descriptive utility**.

More specifically, the following four limitations are established in advance.

First, this paper does not claim that the variable system of subject-injury satisfies thermodynamic equations.

Second, terms such as “entropy,” “free energy,” and “dissipative structure” are used not as precise physical quantities but as auxiliary vocabulary for clarifying structural correspondences.

Third, this paper makes no strong claim that the second law of thermodynamics is directly applicable to society or civilization.

Fourth, the aim of this paper is not prediction but **to make the positional relationship between closure tendencies and points of intervention more legible**.

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## 2. Redesigning the Thermodynamic Analogy

### 2.1 Distinguishing Closure from Homogenization

This paper begins by distinguishing two concepts that are frequently conflated.

**Closure** refers to the condition in which difference inputs capable of destabilizing a frame become harder to flow in — that is, the supply conditions for D (difference-retention) become impoverished.

**Homogenization** refers to the downstream consequence: the phase in which Var(φ) (evaluative diversity) declines and the evaluative space converges.

Closure is therefore a matter of inlet conditions; homogenization is the state change that follows as its consequence.

What this paper seeks to make analogous in thermodynamic terms is not “disorder” in general, but the structure in which **the contraction of difference-input conditions tends to produce the contraction of evaluative space**.

### 2.2 The Entropy Analogy and Evaluative Closure

In physical systems, closed systems tend toward homogenization. When this structure is introduced by analogy into social and institutional contexts, what this paper is claiming is the following.

In institutions and civilizations, when the cost of maintaining D (difference-retention) is designed to exceed the cost of eliminating difference, the system tends toward evaluative closure.

“Evaluative closure” as used here is not physical isolation. It is the informational, institutional, and evaluative closure in which alternatives, dissent, questions about premises, and external perspectives effectively cease to flow in.

The entropy analogy in this paper is not a moral narrative — “society breaks down if left alone.” Rather, it is a conditional structural description: **in designs that impose high costs on the side that maintains difference, closure tends to be selected**.

### 2.3 Open Systems and Informational Openness

In thermodynamics, order maintenance often requires open systems. If this paper borrows from that intuition, it is only in the following respect.

For subjects and institutions to maintain recomputability, the inflow of difference inputs from outside — that is, the inflow of D — is necessary. In this sense, subjects and institutions require openness in an informational and evaluative sense.

The openness referred to here is not "accepting everything." Rather, it is **not eliminating all inputs capable of betraying the existing frame**.

As Child Paper 3 showed, asymmetry in the relational field can produce the relational outsourcing of recomputation even without explicit coercion. As Child Paper 5 showed, institutions can advance closure through the designed simplification of evaluative space. This paper repositions these as variations of the same tendency — a tendency toward the impoverishment of difference exchange with the outside — and places them on the same schematic map.

### 2.4 The Free-Energy Analogy and E_r_eff

E_r_eff (effective recomputation energy), introduced in Child Paper 2, was the available resource enabling a subject to perform the high-cost operation of recomputation. In thermodynamic vocabulary, this can be read as a concept analogous to "available capacity to do work."

Here too, no strict isomorphism is claimed. What this paper claims is only a structural correspondence: in a system where E_r_eff is constrained, the low-cost elimination of difference or reversion to an existing frame tends to be more stable than the high-cost operation of recomputation.

What is important is that E_r_eff is **not a fatalistic variable beyond intervention**. The supply conditions for D, designs that prioritize speed, H (suspension capacity), and the plurality of evaluative space are all subject to change through institutional design.

This paper is therefore not claiming that "transition to a low-energy state is natural." What it is claiming is that **the cost structure under which such transitions tend to be selected is itself changeable**.

### 2.5 How to Use the Dissipative Structure Analogy

Any use of the term "dissipative structure" in this paper must be limited in its positioning.

Reference to dissipative structures in this paper is limited exclusively to the purpose of the supplementary clarification of the structural intuition that **openness functions as a condition for maintaining order**. No implications beyond this single point are carried in this paper.

The utility of this term lies in its capacity to make visible the following reversal: not "close and stabilize," but "it is precisely because the system is open that recomputability can be maintained." When the four principles of subject protection are read not as mere ethical norms but as **design conditions for keeping the system in an open state**, this supplementary concept functions as an intuitive scaffold.

However, articulating how openness is to be designed is not the role of this paper — that is handled by the four principles of Child Paper 5. The development of non-equilibrium thermodynamics beyond the dissipative structure analogy is also outside the scope of this paper.

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## 3. Closure Tendencies and Points of Intervention

### 3.1 Three Scales of Closure Tendency

What this paper aims to show through the thermodynamic analogy is that closure tendencies appear at different scales as different cost arguments.

**At the individual scale**, when maintaining and re-evaluating difference itself becomes high-cost, subjects tend to select difference elimination. This corresponds to the depletion of E_r_eff described in Child Paper 2.

**At the institutional scale**, simplifying the evaluative space and prioritizing comparability, manageability, and speed reduces operational cost; institutions therefore tend to tilt toward eliminating difference. This corresponds to the second pathway described in Child Paper 5.

**At the civilizational scale**, standardization, efficiency, and optimization reduce the costs of broad coordination; homogenization therefore tends to be selected over difference-retention. It is at this scale that the structure of the entire civilization "tilting toward closure" first becomes visible.

What these three scales have in common is the design condition under which eliminating difference costs less than retaining it.

### 3.2 A Map of Points of Intervention

| Scale | Mechanism of closure | Points of intervention | Primary connections |
|---|---|---|---|
| Individual | Difference elimination due to depletion of E_r_eff | D supply circuit; securing H (suspension capacity) | Child Papers 2 and 3 |
| Institutional | Designed simplification of evaluative space | Implementation of the four principles | Child Paper 5 |
| Civilizational | Closure through cumulative efficiency and standardization | Institutional protection of evaluative-axis plurality | This paper; Chapter 7 of parent paper |

**Closure tendencies exist. But they are not an inevitable fate beyond intervention — they are design tendencies with identifiable points of intervention.**

### 3.3 The Maintenance Dynamics of Closure

Closure does not occur simply because it is low-cost. It is also actively maintained by multiple dynamics. A map of points of intervention that does not account for these maintenance dynamics remains superficial.

**Vested interests and comparability:** There are parties who benefit from the simplification of the evaluative space. Concentration on measurable indicators not only reduces operational costs but also stabilizes the authority of evaluators. Closure is not a design failure — it is also a rational choice for certain parties.

**Avoidance of responsibility and prioritization of speed:** Retaining difference and subjecting it to re-evaluation requires time. In environments that prioritize speed and efficiency, suspension and deliberation are treated as costs. Eliminating H (suspension capacity) is in many cases justified as “productivity improvement.”

**Inertia and friction:** An evaluative space that has once been simplified produces a group of subjects who have adapted to it. Changes in design can mean disadvantage for that group. Closure is self-reinforcing — a state can arise in which the cost of changing exceeds the cost of maintaining the current state.

Intervention therefore involves not merely the addition of difference but the disruption of the incentive structure that maintains closure. The four principles of subject protection (Child Paper 5) can be read not only as design conditions for openness but as counter-conditions that alter the cost structure maintaining closure.

### 3.4 The Continuity of Design Responsibility

The closure tendencies described in this paper do not imply the absence of responsible agents. They are understood as the cumulative result of design choices at the individual, institutional, and civilizational scales that have made difference elimination lower-cost than difference-retention.

At the individual scale, it is relatively possible to identify who suppressed difference and who did not permit suspension. At the institutional scale, responsibility is distributed across operational design, evaluative design, and support design. At the civilizational scale, identifying a single agent of harm is difficult — but this does not mean that responsibility disappears. There, the unit of responsibility is the accumulation of collective design choices.

The phrasing “a structure that is no one’s fault” frequently obscures “the accumulation of designs that someone made.” This paper does not use the thermodynamic analogy for that obscuration. Rather, the reverse: it uses it to show that **the cost structure under which closure is preferred is itself a design object**.

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## 4. The Scope and Limits of the Analogy

### 4.1 What This Paper Deliberately Excludes

Every theoretical framework foregrounds some things and backgrounds others. The thermodynamic framework adopted in this paper is no exception.

First, this framework tends to background **the intentionality of agents**. Local distinctions — good intent or malicious intent, support or domination — recede behind the structural description of closure tendency. Child Paper 3 compensates for this.

Second, this framework tends to weaken the **immediacy of short-term intervention**. A description of structural tendencies can diminish the urgency of "what must be stopped here and now." The design principles of Child Paper 5 compensate for this.

Third, this framework tends to weaken **the intensity of ethical response**. Anger, a sense of responsibility, and the felt sense of injury are easily neutralized within structural description. Child Paper 1's conditions for the use of vocabulary, and the subject protection argument of the parent paper as a whole, compensate for this.

The thermodynamic framework of this paper is therefore not self-contained. It is an auxiliary descriptive device that plays a limited role within the series.

**Conditions of use for the practitioner:** The schematic map in this paper presupposes that its users subject their own evaluative axes and the range of difference inputs they hold to self-audit. This paper therefore does not presuppose an unexamined externality on the part of the practitioner. Recognizing that those who use the map may themselves be standing within closure tendencies is the honest condition for using this paper.

### 4.2 The Legitimacy of Descriptive Scaffolding

That said, the reasons for using this framework are clear.

First, it makes the common structure of closure tendency across individual, institutional, and civilizational scales more legible. Second, it makes clear that maintaining difference-retention and recomputability is not "static preservation" but "the design of openness conditions." Third, it makes it possible to reread the argument for subject protection not as a mere ethical demand but as an intervention design against closure tendency.

This descriptive scaffolding remains open to revision if the proposed closure tendencies fail to correspond to observable patterns of evaluative simplification, difference reduction, and intervention effects across scales.

### 4.3 The Limits of Civilizational-Scale Region B

Civilizational-scale Region B — apparently normal, internally halted — is not a claim that civilization is to be ontologically reified as a single subject. It is a term for operationally describing the state in which, across the totality of multiple institutions, **the maintenance of surface functioning and the decline of the capacity for frame-level renewal coexist**.

This description does not imply the anthropomorphization of civilization as having a will, emotions, or self-identity. Nor does it carry any eschatological implication that "civilization is declining." There is only one intended purpose: to make it possible to operationally describe the question opened by Chapter 7 of the parent paper — "can civilization as a whole lose L3-level renewal capacity while maintaining high output?" — as an analytical construct used in that limited sense.

For example, a society may continue to exhibit high productivity and coordination while losing the practical capacity to reorganize its evaluative axes across education, governance, and public discourse. This is the kind of condition the construct is designed to describe — not a verdict on civilization, but a pattern of evaluative convergence that persists alongside surface stability.

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## 5. Connection to the Companion Papers and the Parent Paper

This paper, as the terminal paper of the series, completes the line of connection to the civilizational scale.

In connection with Child Paper 2, E_r_eff is reread as a resource condition for which intervention in closure tendencies is possible. In connection with Child Paper 3, the relational field is read as the point of divergence between maintained openness and unidirectional closure. In connection with Child Paper 5, the four principles are read as institutional design conditions for keeping the system in an open state. In connection with Chapter 7 of the parent paper, civilizational-scale Region B is redescribed not as inevitable decline but as **a progression of closure that design change can prevent**.

The role of this paper is clear.

> **This paper redescribes subject-injury as closure tendency and provides, as a schematic map, the maintenance dynamics and points of intervention across three scales: individual, institutional, and civilizational.**

This is the distinctive contribution of Child Paper 4 as a standalone paper. Child Paper 2 handles the dynamics of "why the subject stops." Child Paper 3 handles "why relationships become a medium of injury." Child Paper 5 handles "why subject-injury is reproduced at scale." This paper adds to these the schematic map of "why the tendency toward difference-reduction is structurally selected, and where intervention is possible."

Closure tendencies need not presuppose a single intentional designer; they can emerge as the cumulative outcome of many local design choices that repeatedly make difference-elimination lower-cost than difference-retention. The unit of design responsibility is therefore not a single agent but the accumulated structure of those choices — and the task of intervention is to alter that structure.

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## Author’s Note on AI-Assisted Research and Writing

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